Experiment 3: Information Theory

Title

Entropy and Information Gain Calculation using the Play Tennis Dataset

Objective

To calculate the entropy of the dataset, compute the information gain of each input attribute, and identify the attribute with the highest information gain.

Dataset Description

The Play Tennis dataset contains 14 records with four input attributes:

- Outlook
- Temperature
- Humidity
- Wind

The target attribute is Play, which contains two classes:

- Yes
- No

Step 1: Load the Dataset

The Play Tennis dataset was loaded from the CSV file using the Pandas library.

Step 2: Calculate Entropy

The entropy of the target attribute (Play) was calculated to measure the uncertainty in the dataset.

Entropy = 0.9403

Step 3: Calculate Information Gain

Information Gain was computed for each attribute by subtracting the weighted entropy after splitting from the total entropy.

The obtained values were:

- Outlook = 0.2467

- Temperature = 0.0292
- Humidity = 0.1518
- Wind = 0.0481

Step 4: Select the Best Attribute

The attribute with the highest Information Gain was selected as the root node for the Decision Tree.

Outlook had the highest Information Gain (0.2467).

Analysis

The Outlook attribute reduces uncertainty more than the other attributes. Therefore, it provides the most useful information for predicting whether tennis can be played. Temperature has the lowest Information Gain, indicating that it contributes the least to the classification process.

Conclusion

The entropy and information gain of all attributes were successfully calculated. Outlook was identified as the best attribute for the first split in the Decision Tree because it has the highest Information Gain.